# When a longer route buys time: fixed escape plans for regional mine-flood alarms

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**Abstract**

A mine-flood alarm may identify a region without locating the water source or revealing how long the flood has been spreading. Can a route chosen before the accident still get a worker out after a delayed warning? We study one fixed route for each starting point and reported region. The route must reach an exit for every candidate source in that region; distance is minimized only after this requirement is met. On a tunnel network extracted from a real mine plan, we evaluate all 1,587 candidate source nodes, all 1,587 starting nodes, four alarm regions, and three delay targets. For a northwest alarm and a 30-minute delay, rerouting increases the number of covered outside-region starting points from 720 of 1,243 to 1,200. A 50% detour allowance covers 961, so much of the gain requires substantial extra walking. Gains occur in all four regions, but differ sharply in size and cost. They also depend on which water sources and inflow rates the plan is designed to withstand. An exact finite-scenario search separates cases that rerouting can solve from those requiring better source information or earlier departure. These are guarantees within an uncalibrated flood model, not field-validated safety predictions. The study shows how to assess a regional escape plan by the warning delay it can tolerate and the walking distance that tolerance costs.

## 1 Introduction

A worker receives an alarm: water has entered the eastern part of the mine. The shortest way to a shaft runs through that area. Another route is longer, but might remain open for longer. The worker does not know the exact source of the water and cannot tell how much time was lost before the alarm arrived. Which route should the emergency plan recommend?

This question differs from finding the fastest route for a known flood. Water travels through the same tunnels that people use to escape. A route that is short on a dry map may pass through a tunnel that closes before the worker reaches it. A route calculated for one water source may fail if the source is somewhere else in the reported region. For a plan to be usable with a regional alarm, its recommendation must work without that missing information.

We ask how much extra distance is needed to withstand a specified notification delay. For each starting point and alarm region, we choose a single route in advance. The worker follows it immediately on receiving the alarm. A 30-minute guarantee means that the route remains feasible if departure occurs at any time up to 30 minutes after the inrush begins, and that the worker can then complete the journey. It does not mean that the journey takes at most 30 minutes, or that the worker should wait.

The main finding is that this choice can matter across whole parts of a mine network. In the CAD-derived network studied here, a northwest alarm leaves 523 of 1,243 outside-region starting points without a 30-minute guarantee on their dry shortest routes. Changing the route resolves this for 480 of them. The gain persists at 10- and 20-minute targets. It is not always cheap: allowing only 20% extra distance recovers 92 of the 480 points, whereas removing the distance cap recovers all 480. Other regions show smaller gains or require much longer detours.

These results answer a planning question, not a hydraulic one. We use a volume-conserving flood model inherited from a modeling-competition study. The mine geometry comes from a CAD plan, but the flood parameters have not been calibrated to that mine. The contribution is to formulate the information available to the worker, compute the shortest route that meets a regional delay guarantee, and establish where that guarantee is valuable, costly, or impossible in the resulting scenarios. We do not claim a new general shortest-path algorithm or an operationally validated evacuation system.

**Relation to earlier work.** Mine-inrush routing has been studied as an earliest-arrival problem on a time-varying network [1]. Sensor-placement work asks how soon an inrush can be detected [2]. Our question connects these two decisions: what should a fixed escape plan prescribe when detection supplies only a source region and may arrive late? The computational ingredients come from established time-dependent routing and robust optimization. In particular, first-in–first-out travel permits arrival-time dominance [3, 4]; selecting among prepared policies after observing uncertainty is the broader setting of $K$-adaptability [5]. Here, the observation is deliberately limited to the reported region.

The study began as a follow-up to a mine-inrush modeling competition [6]. An initial competition-network example suggested a benefit from detours. The main evidence in this paper is the subsequent enumeration of every eligible source and starting node in all four regions of the CAD network. The earlier example is retained as a secondary comparison in Appendix D, rather than used to establish how common the benefit is.

## 2 A mine, a regional alarm, and a fixed route

### 2.1 The network and the information available

The network has 1,592 junctions and 2,084 undirected tunnels, totaling approximately 248 km. Four shaft-bottom nodes are designated escape endpoints. Reaching the surface after reaching a shaft is outside the model. The graph was extracted from an anonymized CAD mine plan by snapping nearby features, splitting intersections, and connecting the shafts. Supplied roof elevations are used as a proxy for relative floor elevations under a constant-height assumption. Figure 1 shows the resulting geometry and the four alarm regions.

The median $x$ and $y$ coordinates divide the network into northwest (NW), southwest (SW), southeast (SE), and northeast (NE) regions. These are convenient, reproducible partitions, not verified sensor sectors. Excluding the four endpoints and one auxiliary reference-source node leaves 1,587 candidate water sources and the same 1,587 worker starting points. Each flood scenario has *one* source, not all sources active at once. For a regional alarm, its source may be any eligible node in that region.

The plan specifies a route $P(v, R)$ for a worker starting at node $v$ when the reported source region is $R$. The route cannot depend on the precise source, which is unreported. It also does not change with subsequent observations. This is a mathematical model of a region-indexed route map, rather than a design or human-factors evaluation of physical signs.

### 2.2 What it means to tolerate a late alarm

Let $t_0$ be the time the inrush starts, $t_{\text{det}}$ the time it is detected, and $t_{\text{rec}}$ the time the worker receives the alarm. With immediate departure, the relevant delay is

$$d = t_{\text{rec}} - t_0 = (t_{\text{det}} - t_0) + (t_{\text{rec}} - t_{\text{det}}).$$

A detection timestamp need not reveal $t_0$: water may have traveled before reaching a sensor. Conversely, a witnessed event might identify onset closely. We do not assume that the true delay

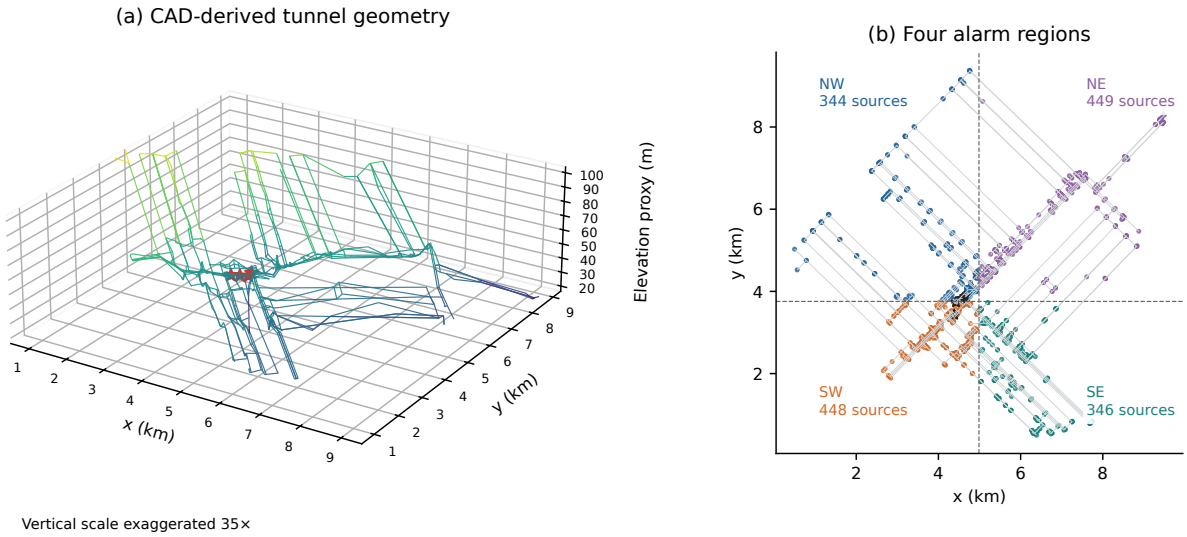



**Figure 1:** The CAD-derived network in three dimensions and in plan view. Stars mark escape endpoints. The vertical scale is exaggerated 35 times; elevations are model inputs, not a surveyed flood surface, and the coordinate-based alarm regions are illustrative.

is always known or always unknown. Any response time before departure can also be included in $d$.

The planner selects a target $\tau$, such as 30 minutes. The route must work for every $d \in [0, \tau]$, whether or not the worker knows the realized delay. This is a requirement against which to assess a plan; the calculation does not establish that an actual alarm system meets it. We also assume that the reported region contains the true source. An alarm that can misclassify the region requires a larger source set.

## 2.3 Flood and walking assumptions

The routing calculation needs to know when each tunnel becomes wet and when it becomes impassable. We generate those times with an event-driven volume-transfer model. Water divides equally among available level or downhill outlets; if none is available, it enters uphill outlets. A shallow advancing front covers a tunnel before it transmits water further. Water accumulates in blocked branches and backs up. Tunnel depth is represented by stored water volume divided by tunnel width and length. Appendix B gives the rules needed to interpret the schedules.

The main experiments use a constant inflow of $0.5\,\mathrm{m}^3/\mathrm{s}$, a width of 4.99 m, a height of 3 m, and a front depth of 0.1 m. A tunnel closes permanently when its mean depth reaches 0.3 m. Walking speeds are 1.3 m/s when dry, 0.7 m/s with the assigned water-flow direction when wet, and 0.5 m/s against it. Speed changes during traversal if the tunnel becomes wet before the worker leaves. These values specify a test model; they are not measured walking laws or a validated human-safety threshold.

Water never recedes in this model. Earlier entry into a tunnel cannot cause later arrival at its far end: this is the first-in–first-out (FIFO) property. Consequently, if a fixed route works when departure occurs at $\tau$, it also works at every earlier departure. We can therefore test an entire delay interval by testing its endpoint. The formal argument and the exact travel-time formula appear in Appendix A.

### 2.4 The quantity we compute

Let $\mathcal{S}_R$ be the finite set of source scenarios for region $R$, and let $L(P)$ be the length of route $P$. For starting point $v$ and target delay $\tau$, define

$$L_R^*(v,\tau) = \min\{L(P) : P \text{ reaches an endpoint from } v \text{ after departure at } \tau \text{ in every } s \in \mathcal{S}_R\}. \tag{1}$$

If no such common route exists, the value is infinity. Let $L_0(v)$ be the shortest dry-map distance to an endpoint. The minimum extra distance needed is

$$\rho_R(v,\tau) = \frac{L_R^*(v,\tau)}{L_0(v)} - 1.$$

Thus a 20% allowance succeeds exactly when $\rho_R(v,\tau) \le 0.20$. The primary objective is escape; minimizing length measures the least walking burden needed to achieve it. A longer route is never required merely to improve an average score.

The baseline follows one deterministically selected dry shortest route. It has no knowledge of the alarm region. Its guarantee is nevertheless checked against exactly the same scenarios as the optimized route. In graphs with nearly equal alternative paths, a small length increase can also represent a better choice among almost tied routes; we do not interpret every small increase as a substantive distance tradeoff.

## 3 How much escape coverage can detours buy?

### 3.1 The gain extends beyond one favorable starting point

Table 1 reports all starting nodes outside each source region at a 30-minute target. An entry counts starting points for which a single prescribed route works for every candidate source in that row. It is not a count of successful individual-source trials. Every starting point is included, whether its baseline is safe, unsafe, or already impossible to improve.

**Table 1:** CAD network, 30-minute target, outside-region origins. Optimized columns impose the stated extra-distance cap; "Any" removes it.

| | | | Origins with a common feasible route | | | | |
|---|---|---|---|---|---|---|---|
| Region | Sources | Origins | Baseline | +20% | +50% | Any | Gain |
| NW | 344 | 1,243 | 720 | 812 | 961 | 1,200 | 480 |
| SW | 448 | 1,139 | 659 | 754 | 760 | 940 | 281 |
| SE | 346 | 1,241 | 1,196 | 1,224 | 1,231 | 1,240 | 44 |
| NE | 449 | 1,138 | 1,075 | 1,079 | 1,092 | 1,110 | 35 |

The largest gain occurs for a northwest alarm. Coverage rises from 720 to 1,200 of 1,243 origins, or from 57.9% to 96.5%. Of the 523 origins whose shortest routes do not meet the target, 480 can meet it by changing route. A southwest alarm also gives a substantial gain: 659 to 940 of 1,139 origins. The southeast and northeast gains are smaller because their shortest-route baselines already cover most outside-region origins.

This is not peculiar to the 30-minute threshold. For northwest alarms, rerouting adds 467 origins at 10 minutes and 479 at 20 minutes; for southwest alarms, it adds 337 and 334. Figure 2 shows these comparisons alongside the distance tradeoff. The number of *additional* origins need not decrease as the delay target increases: both the baseline and the optimized coverage can decrease, at different rates.

The gains are spatially concentrated, not 480 independent replications. The 480 newly served northwest origins form five connected components in their induced subgraph, of sizes 402, 59, 17,

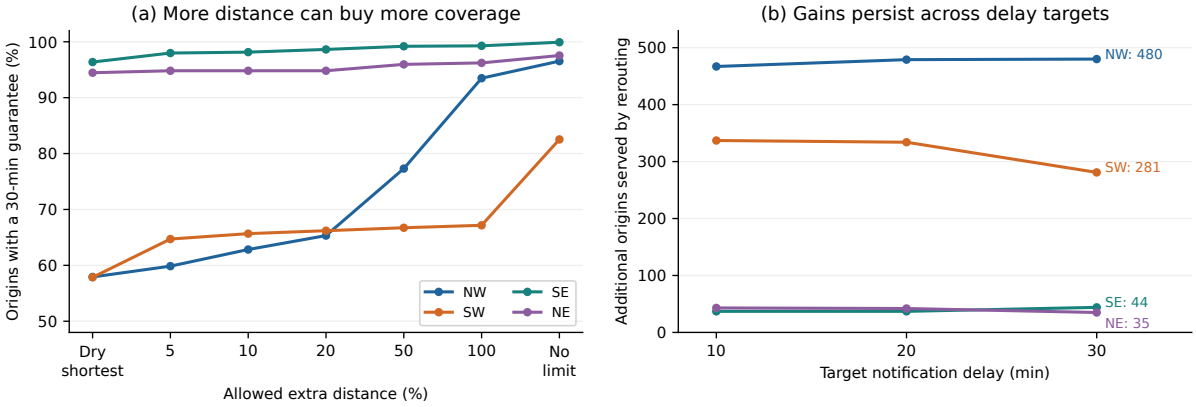



**Figure 2:** The cost and extent of rerouting gains for outside-region origins. Panel (a) fixes a 30-minute target; panel (b) removes the distance cap and counts improvements over the baseline at each delay target.

1, and 1. The southwest gains occupy nine components, the largest containing 203 nodes. These patterns are consistent with changes that protect connected groups of starting points sharing escape corridors. Appendix C maps all four regions. As a check on uneven CAD segmentation, weighting each origin by half the total length of its incident tunnels still raises northwest coverage from 60.0% to 98.8%, and southwest coverage from 52.4% to 86.0%. Those weights are a geometry diagnostic, not a model of where workers are located.

### 3.2 Small detours help, but they do not capture the full benefit

For a northwest alarm, a 20% allowance serves 812 origins; a 50% allowance serves 961; doubling the allowed distance serves 1,162; removing the cap serves 1,200. The median extra distance among the 480 newly served origins is 49.0%. A study restricted to very small detours would therefore miss much of the available improvement.

The southwest result has a different cost. Its newly served origins require a median 267.4% extra distance, meaning a total route about 3.67 times the original length. Increasing the allowance from 100% to unlimited adds another 175 origins. This is useful as a bound on what changing routes could achieve, but a poor reason to recommend unlimited walking in practice. The longest newly feasible outside-region route is about 12.48 km. Fatigue, injuries, congestion, and shaft capacity are not modeled.

The distinction matters for planning. The calculation can show both an economical improvement and a point at which further coverage becomes expensive. It does not prescribe a universal detour budget. A maximum walking distance, travel-time limit, or other operational constraint would have to be added before producing a usable plan.

### 3.3 An example: 254 extra meters change the exit

For a southeast alarm, consider origin P0036 in Figure 3. Its dry shortest route is 6.195 km and ends at P1081. At least one of the 346 candidate source scenarios blocks that route even if the worker departs at onset. A different fixed route ends at P0986 and is 6.449 km long. The extra 254 m, or 4.1%, buys a route that meets the 30-minute target for *all* 346 sources.

The improvement is not a tiny numerical change to the same path. About 6.176 km of the original directed route and 6.430 km of the replacement are not shared. The two routes have similar total lengths because different corridors lead to different shafts. This example was selected after the full experiment to explain the mechanism; the all-origin results, rather than this selected case, establish its prevalence. Both routes still involve substantial walking.

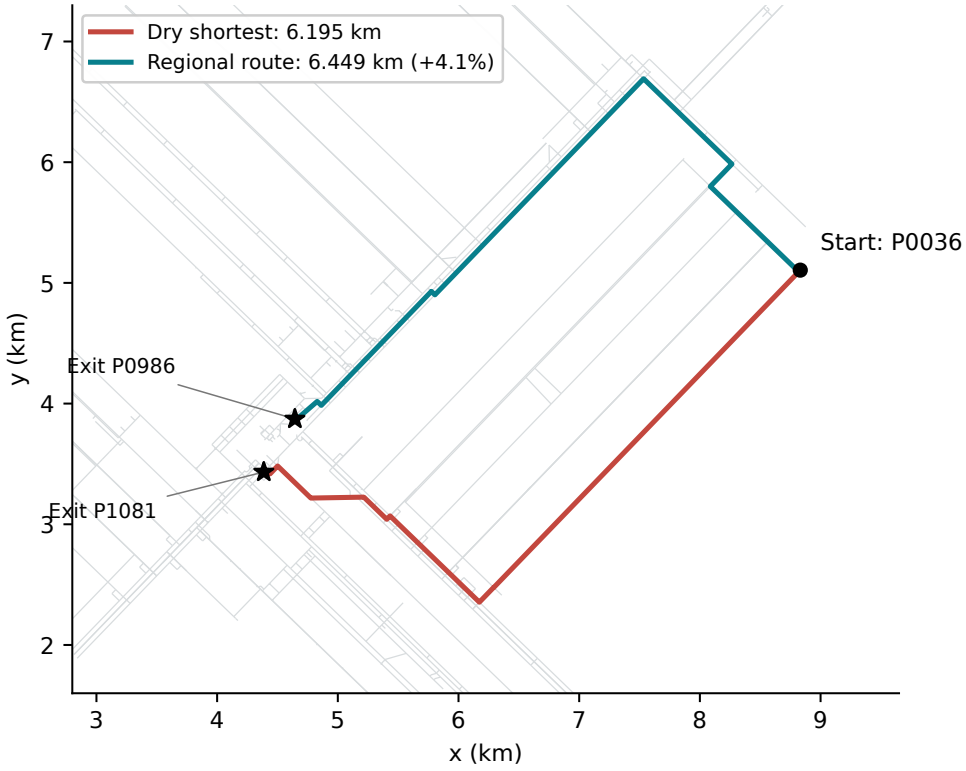



**Figure 3:** Two routes from P0036 under a southeast alarm. The modest difference in total length conceals a large change in corridor and endpoint; the alternative is checked against all 346 regional source nodes.

### 3.4 Some failures require more than a different fixed route

There are two distinct reasons a common route can fail. First, at least one source scenario may leave no feasible route at the target delay, even if the worker is told its precise location. Second, every source scenario may admit an escape route of its own, while no *single* route works for all of them. The latter is a limitation of the available information and the fixed-route requirement.

For southwest alarms at 30 minutes, 199 outside-region origins remain uncovered without a distance cap. Of these, 68 fail even with exact source information in at least one scenario. The other 131 have a route for every individual source but no common route. Better localization could potentially help the second group; it cannot by itself resolve the first. We have not optimized a finer partition or demonstrated how much a real sensor system would recover.

Workers inside the alarm region are also evaluated, rather than discarded. Their results are less favorable: northwest coverage rises from 14 to 97 of 344 origins; southwest from 31 to 139 of 448; southeast from 25 to 138 of 346; northeast from 28 to 179 of 449. Source and worker may coincide in this group. Outside-region coverage must not be presented as whole-mine coverage. Appendix C gives the remaining failure categories.

## 4 How the routes are computed and checked

The difficulty is to find one path that works in every scenario, not to find a different path for each source and then compare the answers. We solve this common-path problem exactly for the supplied finite scenario set, subject to explicit time limits. A timed-out query would be reported as unresolved, never as infeasible.

### 4.1 Search with several arrival times

For a chosen set of source scenarios, a partial route has one length and one arrival time per scenario. At a node, we keep these quantities together as a label. Extending the route along a tunnel updates every arrival time using that scenario's wetting time, closure time, and walking speed. An extension is rejected if it is infeasible in any scenario.

If one label has no greater length and no later arrival in *every* scenario than another label at the same node, the second can be discarded. Any continuation it can use can also be used by the first, because earlier travel does not hurt. Otherwise, both labels are retained. A dry shortest-distance lower bound orders the search, so the first complete feasible route is the shortest common feasible route. The search also discards a label when a reverse, single-scenario calculation shows that it is already too late to reach any endpoint in that scenario.

There is no approximate cap on the number of labels. The method can take exponential time in difficult instances; exactness is not a claim of general computational efficiency. Positive lengths and FIFO travel ensure that an optimal route need not contain cycles.

### 4.2 Checking all sources without using all of them in every search

A region contains hundreds of source scenarios, but not all need to be active in the search from the start. We use finite-scenario constraint generation:

1. Begin with the source that is most restrictive for the baseline route, and find a shortest route feasible for that source.
2. Check the proposed route against *every* source in the region, using backward propagation of the latest feasible departure time.
3. If a source violates the target, add it to the search and solve again. If none does, accept the route. If the search becomes infeasible, report that no common route exists.

This is an exact reduction, not source sampling. At each step, the search solves a relaxation of the full problem. If its shortest route passes the full-source check, no shorter full-source-feasible route can exist. If even the relaxation is infeasible, so is the full problem. Each failed check adds a new source, so the procedure terminates after at most the number of candidate sources, provided the exact subproblems finish. Appendix A states this argument formally.

### 4.3 Experimental scope and verification

The primary study evaluates $1{,}587 \times 4 \times 3 = 19{,}044$ origin–region–target combinations. A restricted-source comparison repeats those combinations, and a doubled-inflow study evaluates all origins and regions at 30 minutes, adding 6,348. All 44,436 queries returned a terminal result; none timed out. Search subproblems had a three-second limit and constraint-generation calls an overall 30-second limit. The source sets, targets, speeds, and main comparisons were recorded before the complete CAD outcomes; the higher-inflow redesign is a subsequent sensitivity study. This is an exploratory computational study, not an independent field confirmation.

We checked the implementation at three levels. On 100 small random graphs with three scenarios each, exhaustive simple-path enumeration agreed with 400 feasibility and minimum-length comparisons; a further 100 maximum-delay brackets contained the enumerated optimum. On CAD subsets, 288 scalar and vector deadline evaluations agreed, and 72 constraint-generation solutions agreed with the full-scenario solver. Finally, five newly feasible routes per region were selected by a deterministic ordering and checked by independent scalar forward traversal against all regional sources, giving 7,935 successful checks. Every accepted nonbaseline route also underwent full-source backward validation during optimization.

The largest relative volume-accounting residual across the 1,587 flood simulations was below $1.02 \times 10^{-13}$. This verifies the accounting, not the realism of the flood propagation. Source

identities, route sequences, binding or infeasibility-witness sources, and solver outcomes are preserved with the analysis.

# 5 What changes when the hazard assumptions change?

## 5.1 A route designed for one inflow may need to change

We first hold the chosen 30-minute routes fixed and double the inflow from 0.5 to 1.0 $m^3/s$. We then redesign the routes for the larger inflow. These answer different questions: whether the original plan survives a change in conditions, and whether a suitable plan still exists.

**Table 2:** Outside-region coverage at 30 minutes after doubling inflow. "Old routes" are frozen at their nominal-flow designs; "Redesigned" solves the problem again at the higher flow.

| | At $Q = 0.5$ | At $Q = 1.0\,m^3/s$ | | |
|---|---|---|---|---|
| Region | Designed routes | Old routes | Dry shortest | Redesigned |
| NW | 1,200 | 855 | 697 | 1,181 |
| SW | 940 | 837 | 632 | 838 |
| SE | 1,240 | 1,235 | 1,194 | 1,238 |
| NE | 1,110 | 1,062 | 1,060 | 1,101 |

The northwest comparison is instructive. Only 855 of the original 1,200 covered origins retain their guarantee under the old route assignments. Redesigning for the higher flow covers 1,181, compared with 697 on dry shortest routes. Much of the loss is therefore a reason to change the plan, rather than evidence that rerouting has ceased to help. For northeast alarms, only two of the 35 nominally newly served origins retain their old-route guarantee, yet redesign still adds 41 origins over the new shortest-route baseline.

In this particular flood model, multiplying constant inflow by a factor divides all flood event times by that factor. The doubled-inflow experiment uses this scaling; a direct rerun at source P0176 reproduced the scaled wetting and closure times exactly in the stored comparison. With slower wet walking and permanent closure, a route feasible at the larger inflow is also feasible at smaller positive inflows, for the same source and model. Thus one can design against an upper inflow bound without requiring the worker to estimate flow during escape. This argument depends on the model's time scaling and does not carry over automatically to pumping, receding water, or different hydraulic dynamics.

## 5.2 A smaller source set can make most of the benefit disappear

The main source set is deliberately broad: every eligible node in the reported region. As a contrasting assumption, we retain only nodes within 50 m of the supplied CAD water-outline polygons. There are 211 such nodes in total. These annotations are not verified source probabilities or a geological risk map; the comparison asks how much the result depends on the hazard set.

Now the shortest-route baselines are already high. Northwest gains fall from 480 to 35 origins; southwest gains from 281 to five; southeast gains disappear. Northeast gains remain, but many require almost no extra length and amount to choosing a nearly tied alternative. The large broad-region gains therefore cannot be advertised without explaining the uncertainty they protect against.

The two source sets are alternative assumptions about the same network, not independent tests of a universal effect. Narrowing a source set can never make the optimal guarantee worse, but it can reduce the *advantage over the baseline*, because the baseline also becomes safer. This distinction explains how a better-informed situation can need less route optimization. Whether

**Table 3:** Restricted-source comparison, with the same outside-region origins and 30-minute target as Table 1. Only source nodes near the CAD water annotations are retained.

| Region | Retained sources | Dry shortest | Redesigned | Gain |
|---|---|---|---|---|
| NW | 47 | 1,204 | 1,239 | 35 |
| SW | 6 | 1,132 | 1,137 | 5 |
| SE | 103 | 1,241 | 1,241 | 0 |
| NE | 55 | 1,097 | 1,137 | 40 |

either source set is defensible for a real alarm requires evidence about geology, sensors, and source localization that is not available here.

## 6 Limitations and implications

The strongest empirical result is a substantial improvement in coverage across several regions of one CAD-derived network. It does not establish the same improvement across independent mines. The four coordinate quadrants are not actual alarm sectors, and enumeration of every source node does not cover every interior point along every tunnel. Escape endpoints themselves are excluded as source candidates. Node counts are not survival probabilities, population estimates, or independent statistical replications.

The main physical limitation is the flood model. It conserves volume but uses equal splitting, a propagation tree, and whole-tunnel mean depths. It does not resolve momentum, depth variation along a passage, junction losses, pumping, debris, or water-velocity effects on a person's stability. Simultaneous-event conventions can affect a propagation-tree model. Geometry extraction, estimated elevations, and shaft connectors also need engineering review. The CAD network supplies a realistic source of geometric complexity; it does not validate the hazard schedules.

The human side is equally limited. Routes terminate at shaft-bottom nodes and ignore evacuation queues, lift capacity, fatigue, injury, and route-following errors. A 12 km mathematical escape route is not automatically an acceptable emergency instruction. The unconstrained results identify the available routing margin before such restrictions are imposed. The next operational step would be to apply defensible walking and access constraints, then measure how much of that margin remains.

Within those limits, the analysis suggests a useful way to assess a plan. First identify which shortest routes fail under the intended alarm information. Then determine whether a different fixed route can meet the delay target, and measure the least extra walking it requires. Finally separate the failures that need better information from those that need an earlier departure even with perfect information. This turns a single coverage percentage into a diagnosis of what should change.

## 7 Conclusion

In the modeled CAD network, allowing a longer route can turn an unsafe shortest-route assignment into a fixed plan that tolerates a delayed regional alarm. The gain can cover connected parts of the network, but its walking cost and sensitivity vary greatly by region. The natural next questions are therefore specific: how much benefit survives calibrated hydraulic schedules and walking limits, and how should alarm regions be chosen to reduce the burden of fixed-route protection? Answering them would move the study from a conditional planning result toward a defensible emergency plan.

**Data, code, and authorship.** The accompanying research directory contains the anonymized network, source definitions, individual route results, solver checks, and figure-generation scripts. The primary CAD audit is in `cad_regional_audit/`; earlier competition results are in `regional_escape/`. No public repository or field-validation dataset is asserted. Research, implementation, and writing received AI assistance. The present follow-up should not be attributed retrospectively to the original competition submission; human authorship review is required before external circulation.

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# A Travel times, guarantees, and exact search

Represent each tunnel by two directed arcs. For arc $a$ in source scenario $s$, let $\ell_a > 0$ be its length, $w_a^s$ its wetting time, $b_a^s$ its closure time, and $v_a^s$ its wet walking speed. Dry speed is $v_d \geq v_a^s > 0$. The uncapped arrival time for entry at time $t$ is

$$\widetilde{A}_a^s(t) = \begin{cases} t + \ell_a / v_d, & \ell_a \leq v_d (w_a^s - t)_+, \\ \max(t, w_a^s) + [\ell_a - v_d (w_a^s - t)_+]/v_a^s, & \text{otherwise}, \end{cases}$$

where $(x)_+ = \max(x, 0)$. Set $A_a^s(t) = \widetilde{A}_a^s(t)$ if it is at most $b_a^s$, and $A_a^s(t) = +\infty$ otherwise. Arrival at the closure threshold is accepted in the numerical convention. A strict safety convention would require an explicit margin.

For a path $P = (a_1, \ldots, a_m)$, compose the arc functions to obtain $A_P^s(t)$. These functions are nondecreasing, including infeasibility as $+\infty$. In particular, earlier traversal of a path cannot turn a feasible journey into an infeasible one. Define its latest feasible nonnegative departure by

$$D_s(P) = \sup\{d \geq 0 : A_P^s(d) < \infty\}, \qquad D_R(P) = \min_{s \in \mathcal{S}_R} D_s(P),$$

with $D_s(P) = -\infty$ if no nonnegative departure works. Backward calculations can also produce negative algebraic deadlines, which diagnose this case; they are not physical departure recommendations.

**Proposition 1** (A single departure-time test)**.** *For the permanent-closure travel schedules above, a path $P$ is feasible in every scenario $s \in \mathcal{S}_R$ for every departure $d \in [0, \tau]$ if and only if $A_P^s(\tau) < \infty$ for every $s \in \mathcal{S}_R$.*

*Proof.* Necessity follows by taking $d = \tau$. For sufficiency, monotonicity gives $A_P^s(d) \leq A_P^s(\tau) < \infty$ for every earlier $d$ and every scenario. □

For an allowed extra-distance fraction $\varepsilon \geq 0$, the complementary maximum-delay question is

$$D_R^*(v, \varepsilon) = \max_{P : L(P) \leq (1+\varepsilon) L_0(v)} D_R(P).$$

With the endpoint convention above, $D_R^*(v, \varepsilon) \geq \tau$ exactly when $L_R^*(v, \tau) \leq (1 + \varepsilon) L_0(v)$. This connects the target-first experiments to the motivating question of how much delay a detour can buy.

**Backward deadlines and impossibility.** For a tunnel with parameters $(\ell, w, b, v_d, v_w)$ and a required downstream deadline $q$, set $e = \min(q, b)$. The latest entry that reaches the far end by $e$ is

$$B(q) = \begin{cases} e - \ell / v_w, & \ell \leq v_w (e - w)_+, \\ \min(e, w) - [\ell - v_w (e - w)_+]/v_d, & \text{otherwise}. \end{cases}$$

Applying this map from the last arc to the first evaluates a fixed path. A reverse label-setting calculation maximizes it over all routes in one known scenario; call the result $U_s(v)$. If $\min_s U_s(v) < \tau$, at least one scenario has no feasible route. If every $U_s(v) \geq \tau$ but the common-path problem is infeasible, the failure is due to incompatible route choices rather than that single-scenario obstruction.

**Proposition 2** (Finite-source constraint generation)**.** *Fix an origin $v$, target $\tau$, and finite scenario set $\mathcal{S}_R$. Suppose each search returns an exact shortest common feasible route for its active subset $T \subseteq \mathcal{S}_R$, or certifies that none exists. If a returned route is feasible for all of $\mathcal{S}_R$, it attains $L_R^*(v, \tau)$. If an active-subset search is infeasible, the full problem is infeasible. Adding a violated source after every failed full-set check reaches one of these conclusions in finitely many additions.*

*Proof.* Every full-set-feasible route is feasible for $T$, so the subset optimum is a lower bound on the full optimum. If its route passes all sources, it is also an upper bound, and the values agree. Infeasibility for $T$ excludes feasibility for its superset. A violated source cannot already be in $T$; hence each failed check strictly enlarges $T$, which is contained in the finite set $\mathcal{S}_R$. □

The statement is conditional on exact subproblem completion. Runtime limits do not provide an infeasibility certificate. The production records contain no such unresolved queries. Exactness refers to the combinatorial search; floating-point feasibility checks use tolerances of order $10^{-7}$ seconds or smaller, not a physical safety margin.

**Why a menu indexed only by delay does not extend the guarantee.** Suppose the source set and route constraints stay fixed, and the worker can choose a prepared route using the exact delay. To cover all delays up to $\tau$, the menu must select at $\tau$ some route that works for every source. That route alone works for every earlier delay by the first proposition. A delay-only menu therefore cannot make the maximum guarantee larger, although it can spare walking when an early alarm permits a shorter route. Observations that also narrow the source set, or hazards that violate monotone feasibility, change this argument.

# B Flood schedules and reproducibility

Each tunnel stores volume between zero and its capacity $WH\ell$, where $W = 4.99\,\mathrm{m}$ and $H = 3\,\mathrm{m}$. A newly reached level or downhill tunnel first receives $Wh_0\ell$ with $h_0 = 0.1\,\mathrm{m}$. If its far node is still dry, it becomes a transmitting edge in a propagation tree; if that node is already wet, it stores further water. An uphill tunnel fills from below before transmitting water to a previously dry upper node. At a reached junction, incoming flow is divided equally among available level or downhill outlets (using a $0.005\,\mathrm{m}$ elevation tolerance); uphill outlets are used when none of those remains. A saturated dead end causes its feeding edge to stop transmitting and store water. Source flow trapped behind full tunnels is transferred through them to the nearest node with storage available; overflow after saturation is included in the volume balance.

Events are ordered deterministically by node and edge identities when simultaneous. Flow assignment follows the propagation tree, and stored volume never decreases. Mean-depth closure at $0.3\,\mathrm{m}$ is recorded separately from the $3\,\mathrm{m}$ capacity. Walking becomes wet-speed travel when water first enters the tunnel; the routing model does not resolve a moving front inside it. These rules explain both conservation and the limitations of interpreting the schedules as hydraulic forecasts.

The source list and input hashes are recorded in `cad_regional_audit/design.json`; parameters and the analysis design are in `protocol.json`. The `full_*.jsonl` files contain all 19,044 primary results. The `near_*.jsonl` files contain the restricted-source comparison; `high_flow/` contains the redesigned doubled-flow results. The fixed-route sensitivity is in `stress.json`. Tables and figures are generated from these records by `regional_alarm_paper/make_cad_figures.py`. The accompanying README gives the run order and explains which simulation caches must be regenerated. No independent accident data are used for validation.

# C Within-region origins and spatial concentration

**Table 4:** Origins inside the reported source region, with a 30-minute target and no distance cap. "Known-source failure" means at least one scenario is individually infeasible; "No common route" means all are individually feasible but incompatible.

| Region | Origins | Baseline | Redesigned | Known-source failure | No common route |
|---|---|---|---|---|---|
| NW | 344 | 14 | 97 | 172 | 75 |
| SW | 448 | 31 | 139 | 202 | 107 |
| SE | 346 | 25 | 138 | 105 | 103 |
| NE | 449 | 28 | 179 | 215 | 55 |

Source and worker may coincide in the table above. Figure 4 locates the outside-region gains; adjacent nodes share corridors and should not be treated as independent replications.

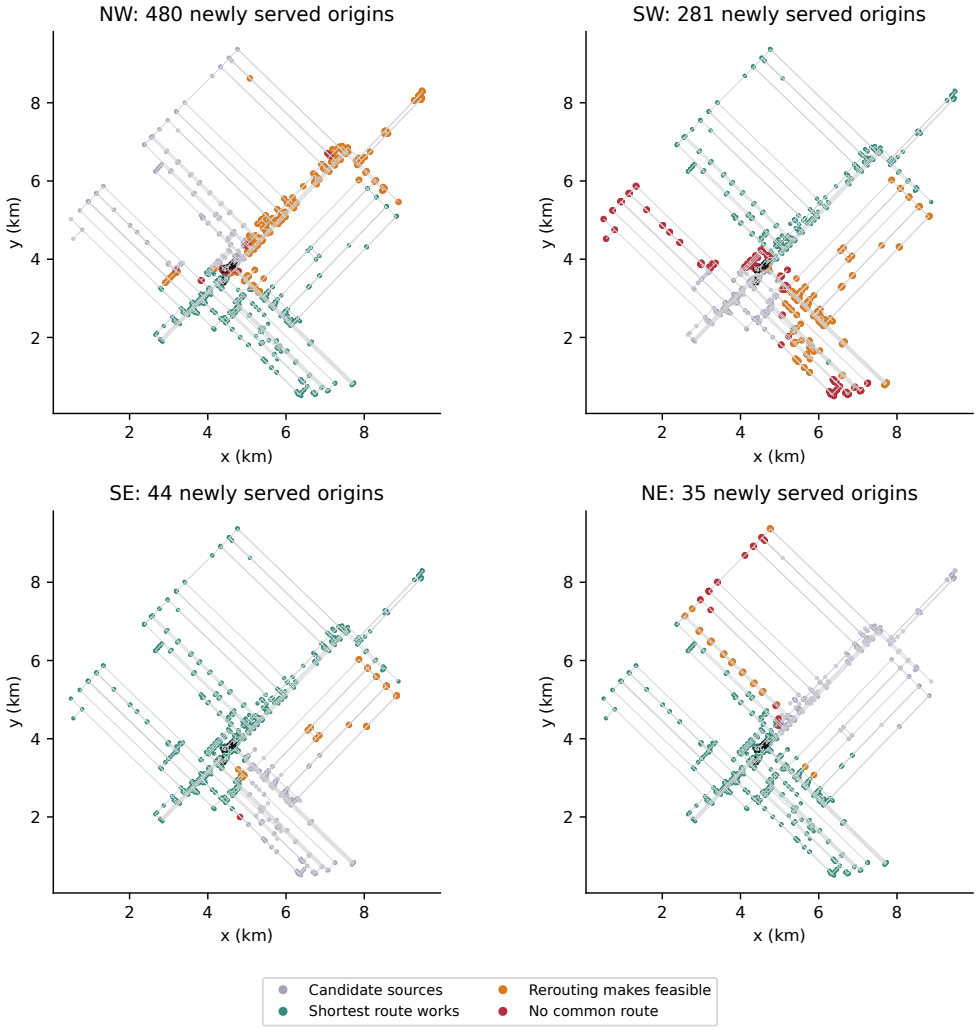



**Figure 4:** All outside-region origins at the 30-minute target. Orange marks points made feasible by rerouting; red marks points for which no common route exists, including those already impossible for at least one known source.

# D The earlier competition-network evidence

An initial screen used two networks, four regions per network, and eight design plus eight held-out sources per region. It selected 123 origin–region pairs for closer study. A promising competition-network case was then expanded to all 113 candidate source nodes in its southeast quadrant. This was adaptive investigation, not a prespecified external confirmation.

With the same slower walking speeds used in this paper, all 548 outside-region origins were evaluated at three delay targets. At 30 minutes, dry shortest routes covered 268 origins, a 20% allowance covered 501, and unlimited distance covered 545. This supports the possibility of a substantial routing benefit in a different graph, but it covers only one region and cannot replace the four-region CAD analysis.

The earlier study also exposed the danger of a small scenario set. A route designed for eight sources and successful on eight held-out sources supported only about 90 seconds when a previously omitted node source was included. Redesigning against the full set repaired that case. Twenty-four additional tunnel-midpoint sources were checked for one selected competition route, but this limited check does not certify continuous source coverage for the CAD study. These results motivate exhaustive *node* enumeration and an explicit statement of what remains outside it.